

Introduction to Information Retrieval

Information Retrieval
Craig Macdonald
2026

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The journey you are searching for:
Glasgow Queen Street (GLQ) to Edinburgh (EDB)
Single Thursday 15 January 2026 • 1 Adult • No Railcard
Edit journey Set journey alert

Please select a journey

Thursday 15 January 2026

View earlier trains Standard from £16.60 First Class from £21.60

This journey will be overtaken by a later departure

12:42 On time

Glasgow Queen Street (GLQ)

Platform 9

→

13:59

Edinburgh (EDB)

Single from £16.60

Select

1h 17m, Direct

12:45 On time

Glasgow Queen Street (GLQ)

Platform 12

→

13:36

Edinburgh (EDB)

Single from £16.60

Select

51m, Direct

Total £0

Continue →
Review tickets and options

Some Characteristics

Structured Query

Structured Data

Accuracy verified

Useful Data is returned

Exact match

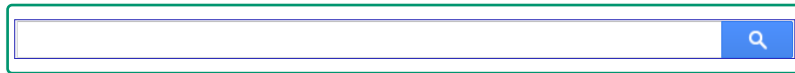
Answer meets query criteria

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Information Retrieval

Information Retrieval is the *science of search engines*



How best to address the **information needs** of users...

- *Effectively*: Get the right information to a user!
- *Efficiently*: Get it to users quickly!

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Information Needs



I want to know what
buildings to see in
Glasgow

~~Glasgow~~ ~~buildings~~ ~~buildings~~

Glasgow Architecture - A Walk About Town - Scotcities.com

www.scotcities.com/central.htm

A look at the fascinating architecture of the landmark buildings in Glasgow's city centre.

Architecture in Glasgow - Wikipedia, the free encyclopedia

https://en.wikipedia.org/wiki/Architecture_in_Glasgow

As a result, Glasgow has an impressive heritage of Victorian architecture: the Glasgow City Chambers; the main building of the University of Glasgow, designed ...

[Glasgow Style](#) - [Victorian era](#) - [Modern era](#) - [References](#)

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Search Engines

- The most visible application of information retrieval technologies are search engines
 - Search engines have evolved since their initial conception 80 years ago



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Search Engines



- Quite effective (at some things)
- Highly visible (some are very widely used)
- Commercially successful (some of them)
 - Google is one of the biggest corporations in the world
 - Underlying technology for searching
- What goes behind the scenes?
 - How do they work?
- Let us have a look at the commercial Search Systems!

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Bing

SEARCH

IMAGES

VIDEOS

MAPS

NEWS

COPILOT

MORE

Isle of Arran

Isle of Arran

Island in Scotland, UK

Overview

Geography

History

Wildlife

Tourism

cottages.com

https://www.cottages.com :

Arran Cottages - Extra Savings End In 5 days

Sponsored Hurry, The Big 2026 Sale just got even bigger, with extra savings on thousands of breaks! Book direct with cottages.com for the best price. Secure your perfect escape today!

Site visitors: Over 10K in the past month

Pet Friendly Cottages

Coastal Cottages

Luxury Scottish Cottages

The Big 2026 Sale

Booking.com

https://www.booking.com :

Places To Stay Isle Of Arran - Browse Our Top 10 Hotels

Sponsored Book your Hotel in Isle of Arran online. No reservation costs. Great rates. Choose From a Wide Range of Properties Which Booking.com Offers. Search Now!

4.5/5 ★★★★★ (88K reviews)

Site visitors: Over 1M in the past month

Top Reviewed Hotels

Homes, Apartments & More

Hotels at Great Prices

Book Now

Macs Adventure

https://www.macsadventure.com , active-holidays , scotland :

Scotland Itineraries - Self-Guided Scotland Tours

Sponsored Explore Scotland independently while we handle logistics. Book your dream trip today!

Explore our collection of self-guided Scotland tours crafted for epic adventures.

On Bing

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Google

Isle of Arran

All Mode

All

Images

News

Videos

Maps

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More

Tools

At Overview



Isle of Arran - Holidays & Breaks | VisitScotland

The Isle of Arran is a place where you can find a little bit of everything you'd ever want from a...
Visit Scotland



Visiting The Scottish Island That Has Everything | Isle of Arran

The Isle of Arran is located off the west coast of...



The Isle of Arran, often called "Scotland in Miniature," is a diverse Scottish island in the Firth of Clyde known for its varied landscapes, from mountainous north to rolling hills offering views over the Firth of Clyde, Glasgow, Galloway, and beyond.

Show more

Isle of Arran

Island in Scotland



Weather

Thu 5° Fri 6° Sat 6°
Google Weather

Directions

13m from Glasgow

Wikipedia

https://en.wikipedia.org/wiki/Isle_of_Arran

Isle of Arran

Arran is an island off the west coast of Scotland. It is the largest island in the Firth of Clyde and the seventh-largest Scottish island, at 432 square kilometres. Historically part of Bute and Arran, it is in the unitary council area of North Ayrshire. Wikipedia

People also ask

Is it worth going to the Isle of Arran?

How do I spend a day on Arran?

How long does it take to drive around the whole of Arran?

What is the Isle of Arran famous for?

VisitArran

https://www.visitarran.com

Welcome to VisitArran

Welcome to the Isle of Arran, Scotland. VisitArran will help you plan and book all the things to do and places to see on your visit throughout the year.

About

The Isle of Arran or simply Arran is an island off the west coast of Scotland. It is the largest island in the Firth of Clyde and the seventh-largest Scottish island, at 432 square kilometres. Historically part of Bute and Arran, it is in the unitary council area of North Ayrshire. Wikipedia

Island groups: British Isles, Islands of the Clyde Area, 432 km²

Population: 4,018

Coordinates: 57°30′29″N 5°14′15″W﻿ / ﻿57.50778°N 5.23750°W﻿ / 57.50778; -5.23750

Highest elevation: Goat Fell 875 m (2,871 ft)

Largest settlement: Lamlash

OS grid reference: NV950220

Feedback

People also search for



Isle of Arran

Isle of Arran

Isle of Arran

Isle of Arran

On Google

What features make Google & Bing different?

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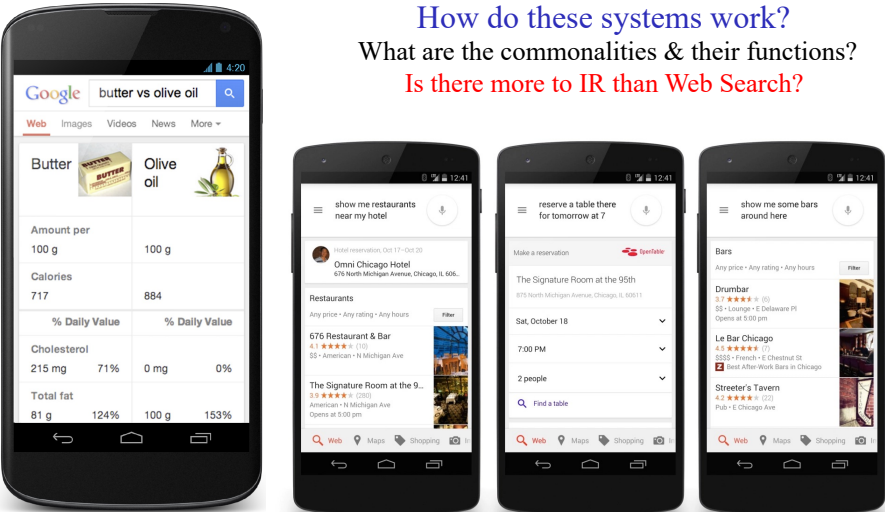
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How do these systems work?

What are the commonalities & their functions?

Is there more to IR than Web Search?



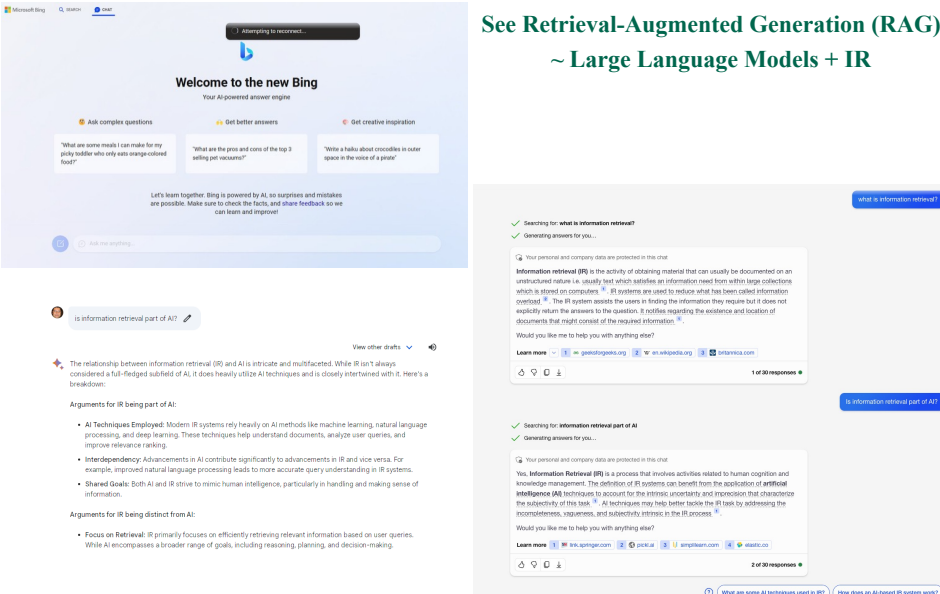
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What's the role of IR in AI Assistants?

The renaissance of IR in the ongoing AI era?

See Retrieval-Augmented Generation (RAG) ~ Large Language Models + IR



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In This Course, We Ask ...

- What makes a system like Google or Bing tick?
 - How does it gather information?
 - What tricks does it use?
 - Expanding beyond the Web?
- How can those approaches be made better?
 - Natural language understanding?
 - Machine learning? Deep learning?
 - User interactions?
- What can we do to make things work quickly?
 - Fast computers? Caching?
 - Compression?
- How do we decide whether it works well?
 - For all queries? For special types of queries?
 - On every collection of information?
- What else can we do with the same approach?
 - Other media? Other tasks?

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Definitions of Information Retrieval

- Salton, 1968
 - Information retrieval is a field concerned with the **structure, analysis, organization, storage, searching and retrieval of information**
- Needham, 1977
 - The **complexity** arises from the **impossibility of describing the content of a document**, or the intent of **request**, precisely, or **unambiguously**
- **General definition**
 - Retrieval of **relevant information** from data sources which were not originally intended for access (e.g. unstructured data)
 - What does this mean?
 - Text (most often - e.g. searching newspaper articles or searching the Web)
 - Images... Video ... Audio....

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Documents vs. Database Records

- Database records (or **tuples** in relational databases) are typically made up of well-defined fields (or **attributes**)
 - e.g. bank records with account numbers, balances, names, addresses, social security numbers, date of birth, etc.
- Easy to compare fields with well-defined semantics to a query in order to find matches

(Unstructured) text is more difficult

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Imprecision in IR

- Most algorithms in Computer Science have a “**right**” answer. In contrast, a **heuristic** tries to guess something close to the right answer.
- Consider the three problems:
 - Sort the following ten integers
 - Find the highest integer
 - Find the beers made by X (i.e. SELECT ... FROM ...WHERE ...)
- Now consider:
 - Find the documents most **relevant** to “hippos in the zoo”.

IR techniques are essentially heuristics because we do not know the right answer

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What Makes a Document Relevant?

- It contains the query terms?
- It contains **ALL** of the query terms?
- It contains the query terms **many times**? (...but what if it is a long document?)
- It contains the query terms many times in a short document?
- It contains the query terms **close** together?
- It is **fresh/recent**?
- It contains terms **similar** to the query terms?
- It is **authoritative** (has many links)?
- It doesn't contain too many ads?
- It doesn't contain too many different, unrelated words (e.g. **spam**)
- It has been **clicked** by many other people for the same query?

ALL of these are heuristics. They are not guaranteed to get a correct, relevant document for all users

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Why to Retrieve?

- Task type ?**
- I need to find some information
 - Who is the head of college of science & engineering?
 - How to get to the LUX city centre from LUX airport?
 - *What is the upcoming topic in IR research?*
 - *What to do this weekend in Glasgow?*
 - Where to search?
 - Newspaper articles, Web pages, Scholarly materials (ACM/IEEE Digital Library), Emails, Tweets, ..., Images (Flickr), ..., Videos (YouTube)
 - ... and many more? Including a combination of all in your own desktop, ... or in your enterprise.... or external sources like the Web!

Time available varies

Exact Need

Vague need

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What do We Mean by “Information”?

- How is it different from “Data”?
 - Information is data in context
 - Databases contain data and produce information
 - IR systems contain and provide information
- How is it different from “Knowledge”?
 - Knowledge is a basis for making decisions
 - Many “knowledge bases” contain decision rules



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What Do We Mean by “Retrieval”?

- Find something that you are looking for:
 - **Ad hoc** search
 - Find documents “about this” topic-x
 - **Known item** search
 - Find the University of Glasgow home page
 - **Answer** seeking
 - What is the capital of Belgium?
 - Directed **exploration**
 - Who makes video conferencing systems?
 - **Decision** making
 - Best places to stay in Paris
 - **Expert** search
 - Who knows about Stable Marriage algorithms in my organisation?
 - Etc.

Role of Interaction

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Scenarios & Applications

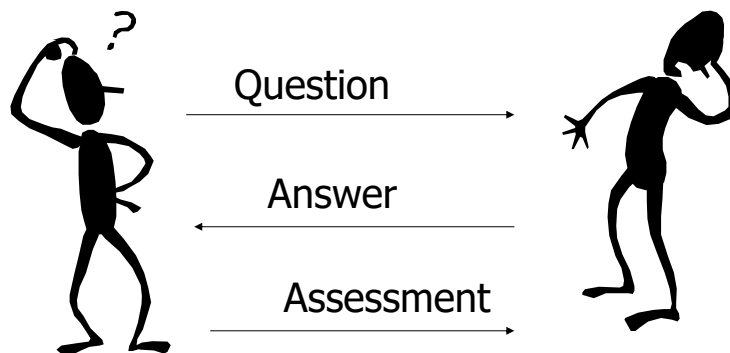
- Web search was a “killer app”
 - Developing advertising models on the web
- Today there are many retrieval applications:



- These types of content share common properties:
 - Text content + some metadata (e.g. title, author, date for papers; subject, sender, destination for email)

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A Question-answer scenario



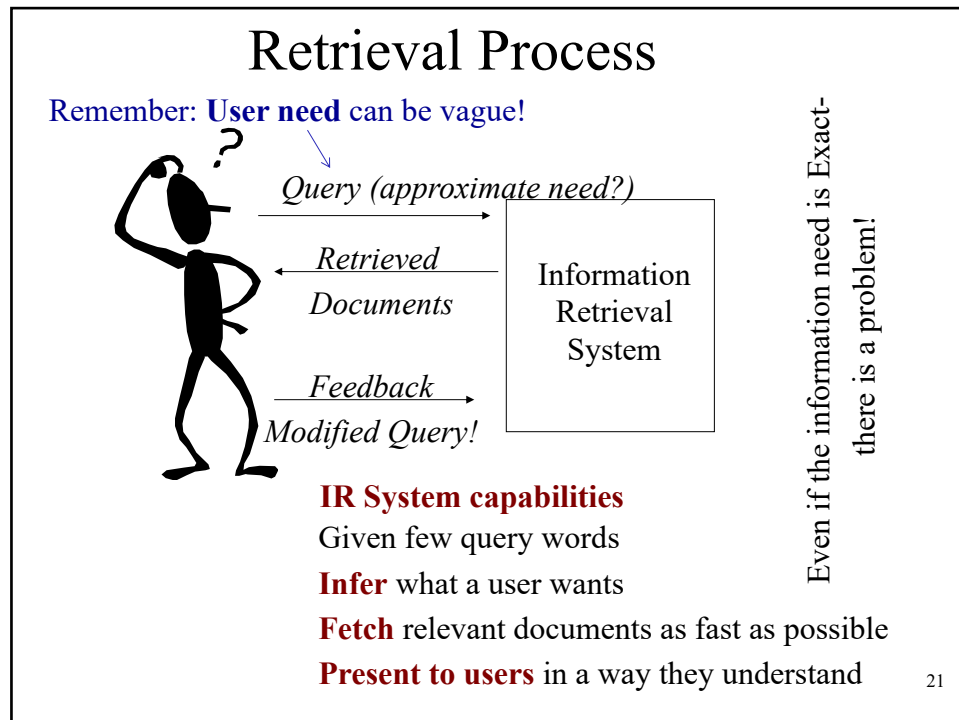
Note 1: Asking a good question can be as hard as answering it

Note 2: The objective of the search engineer is to automate the above process

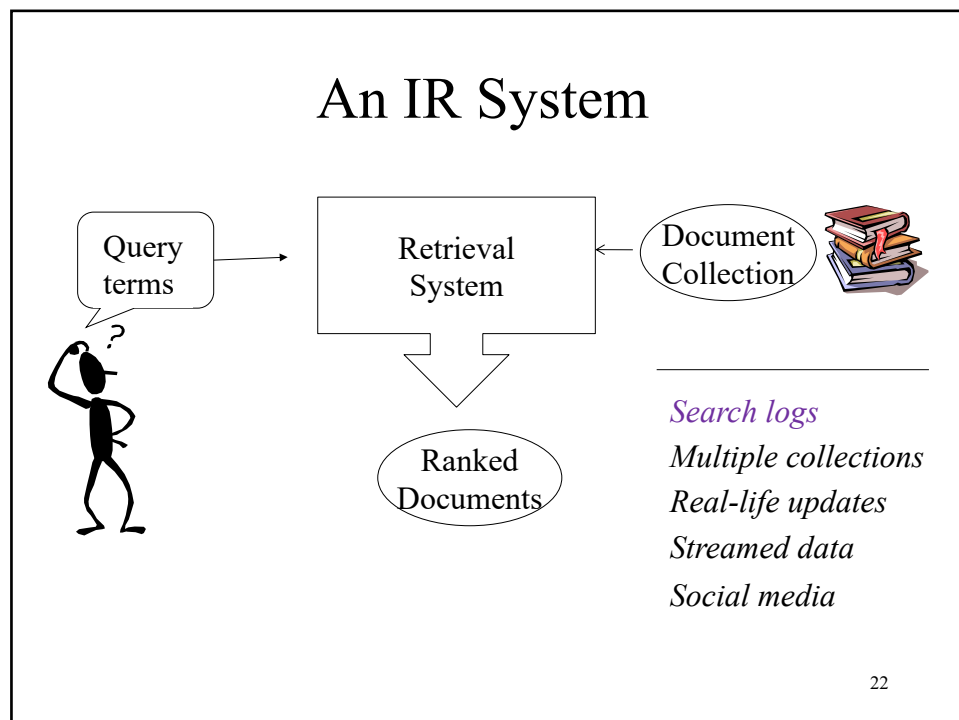
Search/IR Engineers are increasingly sought in industry

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Relevance ... Effectiveness ... Efficiency

- Relevance
 - If the query and the document are **about** the same topic (**known as the Topical Relevance**) – Remember that a user's query can be vague!!
- Effectiveness
 - Looks into the **quality** of the retrieved set. Do they largely contain **relevant** documents? (**role of Retrieval Models**)
- Efficiency
 - Return results as **fast** as possible (**role of search engine architecture - indexes**) – architecture of the IR systems: distributing computation, updating indexes, etc.

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IR H/M: General Information

- Structure
 - In-person lectures: Fridays 10:00-12:00 & [14:00-15:00]
 - Labs: BO1028 (cohort split into 2 groups) on Fridays 14:00-15:00 & 15:00-16:00 - Schedule of lab sessions will be announced on Moodle
- Assessment
 - Assessed coursework (25%), In-class quizzes (5%) & Final exams (80%)
 - Labs: Lab1 (Unassessed); Ex1 (13%), Ex2 (12%),
- Lecturer (s):

Craig Macdonald (Email: craig.macdonald@glasgow.ac.uk)

Sean MacAvaney (Email: sean.macavaney@glasgow.ac.uk)



Craig



Sean

 @craig_macdonald
@macavaney

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Objectives

- At the end of this course you will be able to...
 - Explain the process of retrieval
 - Build an IR system and deploy it for practical applications
 - You may need a retrieval component in your own project work
 - Explain how IR systems are evaluated
 - Understand the web search engine architecture
 - Explain advanced IR technologies & Applications
 - Learning to rank; dense retrieval/neural IR; RAG, personalisation,
- Empower you with a necessary set of skills
 - Develop and deploy IR systems or related technologies

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Planned Syllabus

Fundamental Topics

- Wk 1** - Starting on Monday 12th January - Introduction and IR Systems Architecture
- Wk 2** - Starting on Monday 19th January - Term Weighting Models (Vector Space)
- Wk 3** - Starting on Monday 26th January - Evaluation of IR Systems (Part 1); PyTerrier IR platform
- Wk 4** - Starting on Monday 2nd February - Evaluation (Part 2), Relevance Feedback
- Wk 5** - Starting on Monday 9th February - Advanced Retrieval Models (Probabilistic; Language Models; DFR, etc.)

Advanced/Emerging Topics

- Wk 6** - Starting on Monday 16th February - Advanced IR Models; Machine Learning for IR & Learning to Rank; Web Search (introduction)
- Wk 7** - Starting on Monday 23rd February - Web Search (introduction); Web Search (Link Analysis, Ranking);
- Wk 8** - Starting on Monday 2nd March - Neural Networks for IR, PyTerrier hands-on tutorial on Neural IR
- Wk 9** - Starting on Monday 9th March - Efficient IR (Compression, Pruning, DAAT/TAAT)
- Wk 10** - Starting on Monday 16th March - Web Search Evaluation

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Coursework (25%)

- We will build upon open source libraries
- Deploying, developing and evaluating various retrieval techniques on a medium size Web collection
 - PyTerrier, Pandas, Gensim, Google Colab
- Expose students to a real large-scale IR system
 - Connect concepts from the lectures into code

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Tentative Schedule

Introductory Exercise

	Weight	Handout	Target for Completion
Lab 1	0%	w/c 19th January	Thu 5th February (suggested)

Assessed Exercises

Assessment	Weight	Handout	Due
Exercise 1	13%	w/c 2nd February	Wed 25th February, 4:30pm
Exercise 2	12%	w/c 22nd February	Thu 19th March, 4:30pm

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Quizzes (5%)

- Each week we will have an in-class quiz
- Your score on each of these quizzes will contribute to 5% of your overall grade
- For N quizzes ($N \approx 10$), we will combine your best 8 quiz scores

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Feedback

- Feedback in this course take place in the following forms:
 - 1-1 discussion with lecturers and GTAs in labs
 - Discussions on Moodle
 - Personalised and general feedback on assessed coursework
 - Discussion about quiz answers
 - Office Hours will also be available: c.f. Moodle

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You said, we did

- “Increase weight of coursework”
 - Adjusted exercise weighting (now worth 25%)
 - Added in-class quizzes (worth 5%)

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Teaching Resources

- Text Books – Recommended (available online!)
 - [Search Engines \(Information Retrieval in Practice\)](#), Bruce Croft, Metzer, Strohman, Addison Wesley 2015
 - [Introduction to Information Retrieval](#), Manning, Raghavan, Schutze (eds), Cambridge University Press 2008
- [Course Web page](#) is available on Moodle
 - Updated regularly as we go
 - News/material regarding the course will be posted on Moodle; Interactive quizzes
- Relevant Seminars
 - Mondays, 3-4pm; All are welcome
 - Announced on the school web pages

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First 3 Weeks: Look at the Basics of IR

- Architecture of the System
- Concepts of Relevance & Ranking
- Text Normalisation techniques
- IR Evaluation
- Conducting an IR experiment with PyTerrier

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